

CSCI 04, QUIZ STUDY GUIDE

EVAN DRAKE

```
for i in s:
    print(i)
```

1. QUESTION 2

Recall that the `range(x1, x2)` fn creates a set from `x1` to `x2`. Recall that the `len(x)` fn outputs the amount of elements inside of the collection `x`.

```
s = [42, 51, 23, 1, 2, 3, 4, 5]
for i in range(0, len(s)):
    r = i+42
    print(sqrt(r))
```

2. QUESTION 3

Write some python code with the following:

1. A function with zero inputs that returns zero
2. A function with one input who's output is the input doubled
3. Create a collection of 7 things
4. Give your collection of 7 things to your function from part 2

3. QUESTION 4

Write a python function that does the following:

- Two inputs: `x` and `y` where both `x` and `y` are numbers
- output: x^{y^x}

4. QUESTION 5

Recall that Fermat numbers have the form 2^{2^n} where n is simply a natural number (1, 2, 3, 4, ...).

Write a python function that takes one input n and outputs the Fermat number associated with that input.

5. QUESTION 6

```
for i in s:
    if i > 5:
        print("yes")
    else:
        print("no")
```

6. QUESTION 7

```
def f_squid(x,y):
    if x > y:
        print("Fermat number of x: ", 2**(2**x))
    else:
        print("Fermat number of y: ", 2**(2**y))
```

7. QUESTION 8

```
def not_a_fxn(x):  
    s = x  
    y = s  
    z = "zee or zed, we may never know?"  
    if len(y) > len([]):  
        print(z)  
    elif len(y) == len([1,2,3,4,5]):  
        print("not sure what this does.")  
    else:  
        print("why must I make these?")
```

Email address: drakeev@butte.edu